CareerGraph: Skill-Gap Detection for Early-Career Software Roles

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Abstract

CareerGraph compares a resume with a job posting and lists skills that the posting asks for and the resume does not show. We evaluated it on 1,240 anonymized resumes and 860 job postings collected from January 2022 through December 2023. Top-5 job-match accuracy was 81.4 percent, against 63.2 percent for a keyword baseline. Skill-gap precision was 0.76 and skill-gap recall was 0.69. For data roles the most common missing skill was SQL. For frontend roles the most common missing skill was TypeScript. The false-positive rate on soft skills was 0.18, which is the main weakness of the tool.

1. Introduction

Early-career applicants often apply to roles whose required skills are only partly visible on the resume. A matcher that returns a job title without naming the missing skills is hard to act on. CareerGraph returns both a ranked list of roles and a skill-gap list. This paper measures those two outputs separately. Match accuracy answers whether the role is relevant. Gap precision answers whether a listed missing skill is truly absent from the resume and truly required by the posting.

All labeling was done in Bengaluru by four career coaches who were not the authors. The coaches worked from written guidelines. Each resume-posting pair was labeled by two coaches. A fifth coach broke ties.

2. Data

The resume set contains 1,240 anonymized resumes. Names, phone numbers, and email addresses were removed before labeling. The posting set contains 860 job postings. The collection window is January 2022 through December 2023. Roles are limited to software engineering, data science, and product-adjacent technical jobs. Internships shorter than three months were kept, because they still list skills.

We did not include design-only portfolios or executive biographies. Those documents lack the skill lines CareerGraph expects. A resume with no skills section is still scored. In that case the tool reads projects and experience lines.

3. Method

CareerGraph extracts skill phrases with a dictionary, then embeds the remaining resume sentences and the job description. Similarity is cosine similarity over normalized vectors. The top 5 postings are returned as matches. A skill is reported as a gap only when it appears in the posting and no dictionary alias appears in the resume. Soft skills such as leadership and communication use the same rule, which is why their false-positive rate is

higher.

The keyword baseline counts overlapping skill names and ignores embeddings. It is the only baseline in this paper. We do not claim a comparison with large language models.

4. Results

Top-5 job-match accuracy was 81.4 percent for CareerGraph and 63.2 percent for the keyword baseline. The absolute gain is 18.2 points. Top-1 accuracy was 64.0 percent for CareerGraph and 49.5 percent for the baseline. These match figures use the coach labels as ground truth.

Skill-gap precision was 0.76. Skill-gap recall was 0.69. Precision is higher than recall because the tool stays silent when a skill mention is ambiguous. The most common missing skill among data roles was SQL. The most common missing skill among frontend roles was TypeScript. The most common missing skill among backend roles was Docker, but that third finding is secondary and was not part of the pre-registered hypotheses.

The false-positive rate on soft skills was 0.18. The false-positive rate on technical skills was 0.07. Soft-skill errors usually came from a resume that demonstrated the skill in a bullet without using the dictionary phrase.

5. Limitations

The study covers early-career software roles only. It does not measure interview performance or hiring outcomes. The dictionary is English only. Postings outside the 2022-2023 window were excluded. CareerGraph does not invent skills that are absent from both the resume and the posting.

6. Conclusion

On 1,240 resumes and 860 postings, CareerGraph reached 81.4 percent top-5 match accuracy and 0.76 skill-gap precision. SQL was the most common gap for data roles, and TypeScript was the most common gap for frontend roles. The soft-skill false-positive rate of 0.18 remains the main weakness.

References

[1] Nair, P. and Mehta, A. CareerGraph: Skill-Gap Detection for Early-Career Software Roles. Journal of Applied Career Computing, 7, 2024.